higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL INSTRUMENTS N5

15 August 2024

This marking guideline consists of 6 pages.

Copyright reserved

SECTION A:

QUESTION 1

1.1
- Single hole
- Combined type
- Annubar

(3)

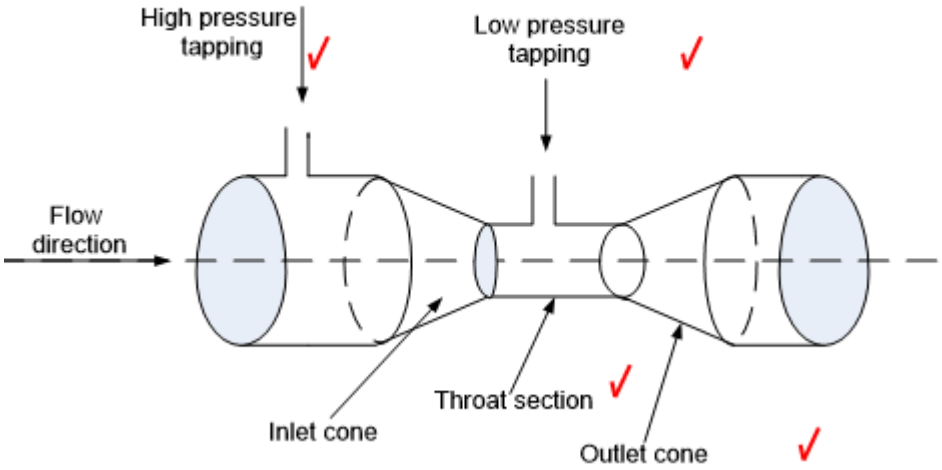
1.2
- The axis of the pitot tube must be parallel to the axis of the pipe
- The fluid shall behave as a single-phase fluid and the velocity of the flow must be between certain limits
- The length of unobstructed straight pipe upstream of the plane measurement should be not less than 20 mm pipe diameter
- The diameter of the head of the pitot static tube must not exceed one twenty-fifth of the diameter

(4)

1.3 1.3.1 It is the irregular movement of a fluid,✓ which is made up of many smaller currents, swirling and moving in all directions. It causes a great deal of frictional resistance to fluid flow.✓ (2)

1.3.2 It is the linear flow of liquid.✓ The control line of the fluid moves the fastest and at the slowest at the pipe wall due to friction.✓ (2)

1.4 1.4.1



(5)

1.4.2 The only factor limiting the size of the meter is the manufacturer. (1)

[17]

QUESTION 2

2.1
- The upstream channel should be relatively wide and deep to assist in uniform velocity distribution.
- The upstream edge of the notch must be square and sharp.
- A free overflow must be maintained.
- The measurement of the head over the crest of the weir or bottom of a V-notch must be made at some distance upstream from the weir plate.
- The underside of the nappe of the weir must be at atmospheric pressure.
- For a suppressed weir, the sides of the channel must be vertical, smooth and in a plane at right angles to the plane containing the weir for some distance upstream and downstream of the weir. (Any 4 × 1) (4)

2.2
2.2.1 Control valve
2.2.2 Vena contracta
2.2.3 Weir
2.2.4 Integrator
2.2.5 Swirl flow meter
2.2.6 Primary flow element

(6 × 1) (6)

2.3
- Rectangular weir
- V-notch weir
- Trapezoidal or Cipoletti
- Parabolic weir
- Suppressed
- Hyperbolic
- Compound (2 × 1) (2)

[12]

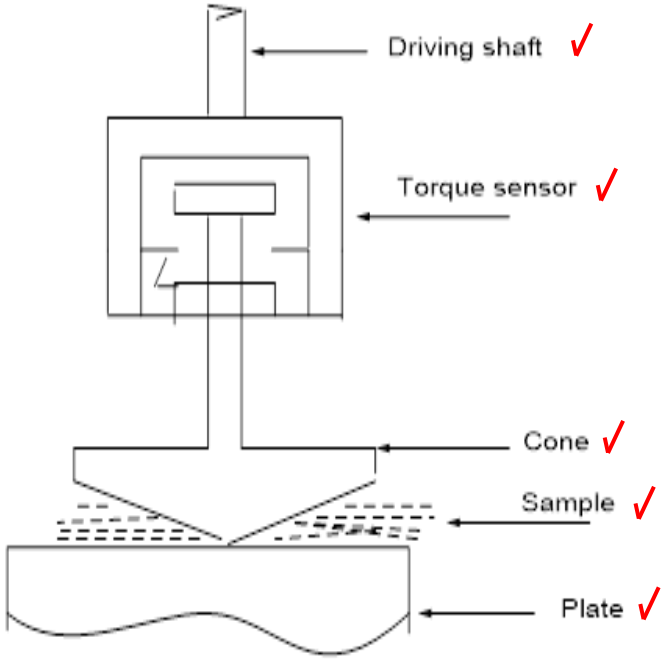
TOTAL SECTION A: 29

SECTION B

QUESTION 3

3.1
- Hand-held hydrometer
- Inductive bridge hydrometer
- Chain-balanced float hydrometer
- Photo electrical hydrometer (4)

3.2



(5)

3.3 3.3.1 Solution conductivity hygrometer or dew cell hydrometer✓✓ (2)

3.3.2
- A – Guard over sensor✓
- B – Taper pipe thread✓
- C – Electrical connection✓ for temperature sensor and power
- D – Coupling✓ (4)

3.4
- Avoid any air bubbles in the sample
- Avoid the use of the an excess amount of fluid around the cone periphery to further minimise the edge effect
- Avoid testing high-viscosity materials such as polymers which tend to ball up and leave a gap between the cone and plate (3)

3.5 3.5.1 Photo-electrical hydrometer (2)

3.5.2
A. Flow in or inlet
B. Hydrometer
C. Flow out or outlet
D. Slit or light gap
E. Photocell
F. Recorder (6)

3.6
- Dirty, corroded, pitted, or deformed spindle
- Off-centre placement of spindle in the container
- Wrong frequency of electric power
- Lack of regular recalibration
- Large air bubbles in the sample
- Turbulent flow or low viscosity fluid with high speed (6)

[32]

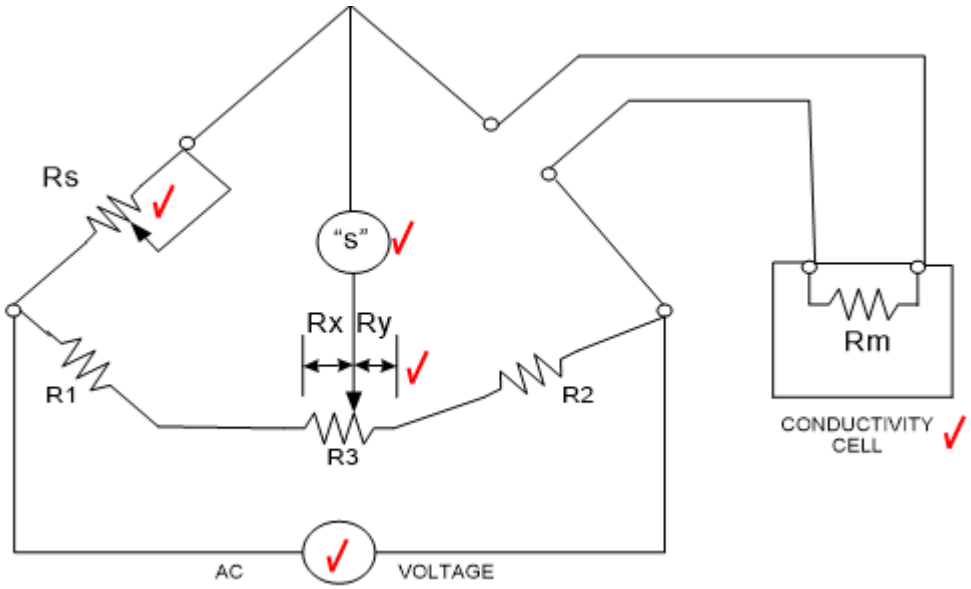
QUESTION 4: pH AND CONDUCTIVITY MEASUREMENT

4.1
- Careful choice of locating the electrode assembly, with a compromise between the greatest distance from the precipitation region and the desirability of close proximity✓to minimise process control lags.
- To incorporate the assembly into the liquid circulating pipeline from the deposition tank.✓
- Bleed arrangement, which is used on effluent treatment plants.✓
- Flow systems can be used in cases where finely divided particles can exert a cleaning action✓ (4)

4.2 4.2.1 Ultrasonic transducer (1)

4.2.2 An ultrasonic transducer is attached to a thin plate.✓ The ultrasonic energy causes the plate to vibrate,✓ thus causing it to shed any materials on its surface.✓ (3)

4.3



(5)

4.4 It is used to nullify the temperature effect on solution conductivity,✓ such that the measurement will always read with respect to some reference temperature✓ regardless of process temperature. (3)

[16]

QUESTION 5: AUTOMATIC CONTROL

5.1
- Position balance
- Force balance (2)

5.2 5.2.1 Proportional control action (2)

5.2.2 The bellows act as error-detecting stages.✓ They determine the difference between the measured variable and the reference or set point.✓ (2)

5.2.3 It is defined as a controller action where the correction (output signal) is proportional to the deviation from the set point. (2)

5.2.4 The controller oscillates around the set point✓ several times before leaving a permanent error called offset.✓ (2)

5.2.5 Integral action must be added.✓ It will integrate any difference between measurement and set point until the difference is zero.✓✓ (3)

5.2.6 Introduce the derivative action (2)

5.2.7
- Should the measured variable (temperature) increase more than the set point, a deviation occurs✓ which causes the flapper to close the nozzle.✓
- This will cause the back pressure on the nozzle to increase,✓ and cause the direct relay valve to increase its output.✓
- The output of the relay will cause the air close valve to restrict the steam flow to the product✓ resulting in the temperature of the product dropping.✓
- The pressure in bellows C will also increase,✓ repositioning the flapper proportionally so that the output is proportional to deviation.✓ (8)

[23]

TOTAL SECTION B: 71
GRAND TOTAL: 100

Copyright reserved